

TECHNICAL REFERENCE · INTERNAL

SMAART MINDS

Database

Documentation

MongoDB Collection Reference — v1.0

PROJECT

SMAART Institute User Dashboard

DATABASE

MongoDB (Mongoose ODM)

PREPARED

27 February 2026

STATUS

Internal · v1.0

COLLECTIONS

8 Collections Documented

TOTAL RECORDS

23 Sample Documents

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1 avatars

Stores each student's gamification avatar state including XP, streak tracking, 3D accessories, animations, and visual customisation options.

2 Records

Ref: users

FIELD	TYPE	DESCRIPTION
<code>_id</code>	ObjectId	Unique document identifier
<code>userId</code>	ObjectId → users	Reference to the owning user
<code>level</code>	Number	Current gamification level (starts at 1)
<code>xp</code>	Number	Experience points accumulated
<code>xpToNextLevel</code>	Number	XP threshold to reach the next level
<code>streakCycleDay</code>	Number	Current day index within the 7-day streak cycle
<code>streakCyclesCompleted</code>	Number	Total fully completed streak cycles
<code>streakStartDate</code>	Date / null	Date the current active streak began
<code>lastStreakDate</code>	String	Last date a streak activity was recorded (YYYY-MM-DD)
<code>streakActive</code>	Boolean	Whether a streak is currently active
<code>streak</code>	Number	Current consecutive streak day count
<code>baseModel</code>	String	URL or identifier of the base 3D avatar model
<code>accessories.shoes</code>	Object	{ unlocked: Boolean, modelUrl: String, equipped: Boolean }
<code>accessories.jacket</code>	Object	{ unlocked: Boolean, modelUrl: String, equipped: Boolean }
<code>accessories.glasses</code>	Object	{ unlocked: Boolean, modelUrl: String, equipped: Boolean }
<code>animations.idle</code>	Object	{ url: String, unlocked: Boolean } — always unlocked by default
<code>animations.celebrate</code>	Object	{ url: String, unlocked: Boolean }
<code>animations.wave</code>	Object	{ url: String, unlocked: Boolean }
<code>animations.dance</code>	Object	{ url: String, unlocked: Boolean }
<code>currentAnimation</code>	String	Name of the currently active animation (default: "idle")
<code>customization.skinTone</code>	String	Hex colour code for skin tone
<code>customization.hairColor</code>	String	Hex colour code for hair colour
<code>customization.eyeColor</code>	String	Hex colour code for eye colour
<code>streakHistory</code>	Array	Historical log of past streak records

FIELD	TYPE	DESCRIPTION
sundayHolidays	Array	Sundays marked as holidays, excluded from streak counting
lastActivityDate	Date	Timestamp of the most recent recorded activity
unlockHistory	Array	Log of all items/animations unlocked over time
createdAt	Date	Document creation timestamp
updatedAt	Date	Last update timestamp
__v	Number	Mongoose internal version key

SAMPLE RECORDS

_ID	USERID	LEVEL	XP	STREAK	STREAKACTIVE	XPTONEXTLEVEL
69a01771...	69a0132c...	1	0	0	false	100
69a01942...	69a0162b...	1	0	1	true	100

2 colleges

Stores institutional information for onboarded colleges including legal documents, contact information, coordinator details, capacity limits, and subscription validity periods.

1 Record

Ref: users (collegeAdmin)

FIELD	TYPE	DESCRIPTION
<code>_id</code>	ObjectId	Unique document identifier
<code>collegeName</code>	String	Name of the institution
<code>institutionType</code>	String	Type of institution (e.g., "College", "University")
<code>governanceType</code>	String	Governance model (e.g., "Private", "Government")
<code>logo</code>	String (URL)	Cloudinary URL for institution logo image
<code>email</code>	String	Official institution email address
<code>website</code>	String	Institution website URL
<code>contactNumber</code>	String	Primary contact phone number
<code>address.street</code>	String	Full street/building address
<code>address.city</code>	String	City
<code>address.state</code>	String	State
<code>address.pincode</code>	String	Postal / ZIP code
<code>address.country</code>	String	Country
<code>onboardingDate</code>	Date	Date the institution was formally onboarded
<code>affiliatedUniversity</code>	String	Name of the parent or affiliated university
<code>coordinatorInfo.name</code>	String	Coordinator full name
<code>coordinatorInfo.designation</code>	String	Coordinator designation/title
<code>coordinatorInfo.email</code>	String	Coordinator email address
<code>coordinatorInfo.mobile</code>	String	Coordinator mobile number
<code>registrationNumber</code>	String	Official institutional registration number
<code>accreditationStatus</code>	String	Accreditation body/grade (e.g., "NAAC")
<code>gstNumber</code>	String	GST registration number
<code>panNumber</code>	String	PAN number
<code>mouDocument</code>	String (URL)	MOU document Cloudinary URL

FIELD	TYPE	DESCRIPTION
universityActProof	String (URL)	University act proof document Cloudinary URL
ndaDocument	String (URL)	NDA document Cloudinary URL
maxFaculty	Number	Maximum allowed faculty capacity
maxStudents	Number	Maximum allowed student capacity
validityStartDate	Date	Subscription/contract start date
validityEndDate	Date	Subscription/contract end date
status	String	"Active" / "Inactive"
numberOfDepartments	Number	Total departments registered
departments	Array	List of department references
totalStudents	Number	Current total enrolled student count
activeCourses	Number	Number of active courses offered
institutionalReports	Array	Generated institutional report records
collegeCode	String	Auto-generated unique college code (e.g., CLG00001)
collegeAdmin	ObjectId → users	Reference to the college admin user account
createdAt	Date	Document creation timestamp
updatedAt	Date	Last update timestamp
__v	Number	Mongoose internal version key

SAMPLE RECORDS

COLLEGECODE	COLLEGENAME	GOVERNANCETYPE	STATUS	MAXSTUDENTS	VALIDITYEND
CLG00001	SRM	Private	Active	1000	2027-02-26

3

counters

Auto-increment sequence tracker used for generating sequential, human-readable IDs for users, teachers, and students across the platform.

3 Records

FIELD	TYPE	DESCRIPTION
<code>_id</code>	String	Counter key name (e.g., "userId", "teacherId", "studentId")
<code>seq</code>	Number	Current sequence value; incremented atomically on each new document creation
<code>__v</code>	Number	Mongoose internal version key

SAMPLE RECORDS

_ID (COUNTER KEY)	SEQ	NEXT GENERATED ID
<code>userId</code>	9	USR000010
<code>teacherId</code>	1	TCH00002
<code>studentId</code>	4	STU00005

4

departments

Stores academic department information scoped to a specific college. Tracks headcount of students and faculty per department.

1 Record

Ref: colleges

FIELD	TYPE	DESCRIPTION
<code>_id</code>	ObjectId	Unique document identifier
<code>name</code>	String	Department full name (e.g., "Computer Application")
<code>description</code>	String	Short description of the department
<code>code</code>	String	Abbreviated department code (e.g., "CA")
<code>college</code>	ObjectId → colleges	Reference to the parent college
<code>studentCount</code>	Number	Number of students enrolled in the department
<code>facultyCount</code>	Number	Number of faculty members in the department
<code>createdAt</code>	Date	Document creation timestamp
<code>__v</code>	Number	Mongoose internal version key

SAMPLE RECORDS

CODE	NAME	COLLEGE	STUDENTCOUNT	FACULTYCOUNT
CA	Computer Application	69a0118d... (SRM)	0	0

5 registrations

Stores comprehensive student self-registration profile submissions including education history, job preferences, work experience, extracurricular activities, and career goals. Records remain "pending" until reviewed by an admin.

[3 Records](#)[Ref: users](#)

FIELD	TYPE	DESCRIPTION
<code>_id</code>	ObjectId	Unique document identifier
<code>userId</code>	ObjectId → users	Linked user account reference
<code>fullName</code>	String	Student's full legal name
<code>email</code>	String	Email address
<code>mobileNumber</code>	String	Primary mobile number
<code>alternateMobile</code>	String	Alternate mobile number (optional)
<code>nickname</code>	String	Preferred display nickname
<code>gender</code>	String	Gender (e.g., "male", "female")
<code>dob</code>	Date	Date of birth
<code>profilePhoto</code>	String (URL)	Cloudinary profile photo URL
<code>institution</code>	String	Name of the institution studied at
<code>department</code>	String	Department name
<code>educationLevel</code>	String	Selected education/sector category
<code>yearOfStudy</code>	String	Year when current study started
<code>yearOfPassing</code>	String	Expected graduation/passing year
<code>sectorPreferences.preferredSectors</code>	Array<String>	Primary preferred career sectors
<code>sectorPreferences.secondarySectors</code>	Array<String>	Secondary/fallback sector preferences
<code>status</code>	String	"pending" / "approved" / "rejected"
<code>currentSessionId</code>	String / null	Current active session ID
<code>lastLogin</code>	Date / null	Most recent login timestamp
<code>previousLogin</code>	Date / null	Previous login timestamp
<code>streakData</code>	Object	Embedded streak state (mirrors avatar streak fields for registration flow)
<code>tenthDetails.schoolName</code>	String	School name for 10th grade

FIELD	TYPE	DESCRIPTION
<code>tenthDetails.yearOfPassing</code>	String	Year of passing 10th grade
<code>tenthDetails.percentage</code>	String	10th grade percentage
<code>tenthDetails.marksheet</code>	String (URL)	10th marksheet document (Cloudinary)
<code>twelfthDetails.schoolName</code>	String	School name for 12th grade
<code>twelfthDetails.stream</code>	String	Stream (e.g., "Science", "Commerce", "Arts")
<code>twelfthDetails.yearOfPassing</code>	String	Year of passing 12th grade
<code>twelfthDetails.percentage</code>	String	12th grade percentage
<code>twelfthDetails.marksheet</code>	String (URL)	12th marksheet document (Cloudinary)
<code>higherEducation</code>	Array<Object>	UG/PG qualification entries. Each: { id, qualificationLevel, degree, specialization, institutionName, university, yearOfPassing, cgpaPercentage, degreeStatus, certificate }
<code>workExperience</code>	Array<Object>	Work/internship records. Each: { id, experienceType, organizationName, jobTitle, industry, startDate, endDate, currentlyWorking }
<code>extracurricular</code>	Array<Object>	Activity entries. Each: { id, activityType, level, description, achievements }
<code>jobPreferences</code>	Array<Object>	Job preference records. Each: { id, preferredRole, jobType, preferredLocation, willingToRelocate, expectedSalary }
<code>projects</code>	Array	Student projects (optional)
<code>certificates</code>	Array	Uploaded achievement certificates (optional)
<code>careerGoals.shortTerm</code>	String	Short-term career goal
<code>careerGoals.mediumTerm</code>	String	Medium-term career goal
<code>careerGoals.longTerm</code>	String	Long-term career goal
<code>personalDevelopmentGoals.shortTerm</code>	String	Short-term personal development goal
<code>personalDevelopmentGoals.mediumTerm</code>	String	Medium-term personal development goal
<code>personalDevelopmentGoals.longTerm</code>	String	Long-term personal development goal

FIELD	TYPE	DESCRIPTION
submissionDate	Date	Date the registration form was submitted
createdAt	Date	Document creation timestamp
updatedAt	Date	Last update timestamp
__v	Number	Mongoose internal version key

SAMPLE RECORDS

FULLNAME	EMAIL	STATUS	INSTITUTION	DEPARTMENT	SUBMISSIONDATE
Sharuk	sharuk@gmail.com	pending	SRM	Computer Application	2026-02-26
Abdul Razith	abdulrazith@gmail.com	pending	SRM	Computer Application	2026-02-26
Kumar	kumar@gmail.com	pending	SRM	KINETIC ENGINEERING	2026-02-26

6 students

Stores core student account records created by college admins. Used for authentication, course enrolment, coaching session management, and assessment tracking within the platform.

3 Records

Ref: colleges, teachers

FIELD	TYPE	DESCRIPTION
<code>_id</code>	ObjectId	Unique document identifier
<code>studentId</code>	String	Sequential human-readable ID (e.g., STU00001)
<code>fullName</code>	String	Student's full name
<code>email</code>	String	Login email address
<code>mobile</code>	String	Mobile number
<code>password</code>	String (bcrypt)	Hashed password (bcrypt, cost factor 10)
<code>role</code>	String	Always "student"
<code>college</code>	ObjectId → colleges	Reference to the associated college
<code>rollNumber</code>	String	Institutional roll number
<code>section</code>	String	Class section (e.g., "C")
<code>department</code>	String	Department name
<code>assignedTeacher</code>	ObjectId → teachers	Reference to the assigned teacher/mentor
<code>enrolledCourses</code>	Array	List of enrolled course references
<code>coachingSessions</code>	Array	Coaching session log entries
<code>status</code>	String	"active" / "inactive"
<code>assessments</code>	Array	Assessment records and results
<code>admissionDate</code>	Date	Date of student admission into the platform
<code>loginHistory</code>	Array	Historical login event log
<code>isFirstLogin</code>	Boolean	Flags whether it is the student's first login
<code>mustChangePassword</code>	Boolean	Forces password change on next login when true
<code>passwordChangedAt</code>	Date	Timestamp of the last password change
<code>lastLogin</code>	Date	Most recent successful login timestamp
<code>lastOtpSentAt</code>	Date	Timestamp of the last OTP dispatch
<code>otpAttempts</code>	Number	Count of OTP verification attempts

FIELD	TYPE	DESCRIPTION
currentSessionId	String (UUID)	Current active session token
isAssessmentCompleted	Boolean	Whether the initial assessment has been completed
isRegistered	Boolean	Whether the student's registration form is fully completed
badges	Array	Earned gamification badges
createdAt	Date	Document creation timestamp
updatedAt	Date	Last update timestamp
__v	Number	Mongoose internal version key

SAMPLE RECORDS

STUDENTID	FULLNAME	EMAIL	DEPARTMENT	STATUS	ISREGISTERED
STU00001	Sharuk	sharuk@gmail.com	Computer Application	active	false
STU00002	Vaasim	vaasim@gmail.com	Computer Application	active	—
STU00003	Kumar	kumar@gmail.com	Computer Application	active	false

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teachers

Stores faculty/teacher account records created by college admins. Teachers are assigned to students and manage classes within the platform.

1 Record

Ref: colleges

FIELD	TYPE	DESCRIPTION
<code>_id</code>	ObjectId	Unique document identifier
<code>teacherId</code>	String	Sequential human-readable ID (e.g., TCH00001)
<code>fullName</code>	String	Teacher's full name
<code>email</code>	String	Login email address
<code>mobile</code>	String	Mobile number
<code>password</code>	String (bcrypt)	Hashed password (bcrypt, cost factor 10)
<code>role</code>	String	Always "teacher"
<code>college</code>	ObjectId → colleges	Reference to the associated college
<code>department</code>	String	Department name
<code>subject</code>	String	Primary subject taught (e.g., "DSA")
<code>status</code>	String	"active" / "inactive"
<code>certificates</code>	Array	Uploaded professional certificates
<code>assignedClasses</code>	Array	References to classes assigned to this teacher
<code>joiningDate</code>	Date	Date the teacher joined the platform
<code>loginHistory</code>	Array	Historical login event log
<code>createdAt</code>	Date	Document creation timestamp
<code>updatedAt</code>	Date	Last update timestamp
<code>__v</code>	Number	Mongoose internal version key

SAMPLE RECORDS

TEACHERID	FULLNAME	EMAIL	DEPARTMENT	SUBJECT	STATUS
<code>TCH00001</code>	Abdul	abdul@gmail.com	Computer Application	DSA	active

8 users

Central identity and authentication collection for all roles on the platform: **admin**, **college_admin**, and **student**. Every person with platform access has a record here. Tracks login history, session state, badges, and profile metadata.

[9 Records](#)[Ref: colleges](#)

FIELD	TYPE	DESCRIPTION
<code>_id</code>	ObjectId	Unique document identifier
<code>userId</code>	String	Sequential human-readable ID (e.g., USR000001)
<code>fullName</code>	String	Full name
<code>email</code>	String	Login email address
<code>mobile</code>	String	Mobile number
<code>password</code>	String (bcrypt)	Hashed password — absent for pending student users
<code>role</code>	String	"admin" / "college_admin" / "student"
<code>status</code>	String	"active" / "pending"
<code>college</code>	ObjectId → colleges	College reference — present for college_admin role
<code>qualification</code>	String	Academic qualification (admin users only)
<code>specialization</code>	String	Specialization area (admin users only)
<code>location</code>	String	Location/city (admin users only)
<code>referralCode</code>	String	Referral code field (admin users only)
<code>certificates</code>	Array	Uploaded certificates
<code>badges</code>	Array	Earned badges (student users only)
<code>profileImage</code>	String (URL)	Profile photo URL (student users only)
<code>activeVisionBoardId</code>	ObjectId / null	Reference to the active vision board (student users only)
<code>currentSessionId</code>	String / null	Current active session token
<code>lastLogin</code>	Date	Timestamp of most recent successful login
<code>lastLoginIp</code>	String	IP address of the most recent login
<code>loginHistory</code>	Array<Object>	Login event log. Each entry: { timestamp, ipAddress, userAgent, browser, os, device, location, success, _id }
<code>createdAt</code>	Date	Document creation timestamp
<code>updatedAt</code>	Date	Last update timestamp

FIELD	TYPE	DESCRIPTION
__v	Number	Mongoose internal version key

SAMPLE RECORDS

USERID	FULLNAME	ROLE	STATUS	EMAIL	LASTLOGIN
USR000001	fareeda	admin	active	fareeda@gmail.com	2026-02-27
USR000002	Ramesh	college_admin	active	ramesh@gmail.com	2026-02-26
USR000003	Sharuk	student	pending	sharuk@gmail.com	—
USR000004	Abdul Razith	student	pending	abdulrazith@gmail.com	—
USR000005	Kumar	student	pending	kumar@gmail.com	—
USR000006	Preethika	admin	active	preethika@gmail.com	2026-02-26
USR000007	Naif Basha	admin	active	naif@gmail.com	—
USR000008	Vishwa	admin	active	vishwa@gmail.com	2026-02-27
USR000009	Rahim	admin	active	rahim@gmail.com	—

9

Collection Relationships

Entity relationships across all collections in the SMAART MINDS database.

```
users (central identity for all roles)
├─ role: "admin"      → Platform-level admin (no college ref)
├─ role: "college_admin" → colleges (colleges.collegeAdmin → users._id)
└─ role: "student"    → registrations (registrations.userId → users._id)
                        ↳ avatars (avatars.userId → users._id)

colleges
↳ departments (departments.college → colleges._id)

students (college-created accounts, separate from users)
├─ college      → colleges._id
└─ assignedTeacher → teachers._id

teachers
└─ college → colleges._id

counters (standalone – auto-increment sequences: userId, teacherId, studentId)
```

RELATIONSHIP	FROM	FIELD	TO
College is managed by admin	colleges	collegeAdmin	users._id
College admin belongs to college	users (college_admin)	college	colleges._id
Student belongs to college	students	college	colleges._id
Teacher belongs to college	teachers	college	colleges._id
Student is assigned a teacher	students	assignedTeacher	teachers._id
Department belongs to college	departments	college	colleges._id
Avatar belongs to user	avatars	userId	users._id
Registration belongs to user	registrations	userId	users._id

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